# RIZVI COLLEGE OF ENGINEERING

**Department of Artificial Intelligence & Data Science**

## Assignment 1, Part (iii): Adapting a Game Dialogue Tool for Marathi

Natural Language Processing (CSDO7011) | Module 1: Introduction | CO1

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| **Assigned NLP application** | Game Dialogue Systems | **Language chosen** | Marathi |

## 1. Why Marathi and What Needs Fixing

I chose Marathi because it is my local language, so I can judge the output myself, and because Tests 6 and 10 in Part (i) already showed how badly Character.AI handles it. In Part (ii), ChatGPT showed that a model can answer in Marathi, but it still drifted between languages and forgot details. So the goal here is an NPC that answers in Marathi, keeps the right level of politeness, knows Maharashtrian culture, and remembers the story. Table 1 turns the problems I observed into the fixes I suggest.

| Problem I observed | Where | Suggested fix |
|---|---|---|
| Replied in English to Hindi, Marathi and Hinglish input | Character.AI, Tests 5, 6, 7, 10 | Language and script detection on the input, a rule to reply in the player's language, and fine-tuning on Marathi dialogue |
| Polite forms ignored (tumhi, Kaka, Bhai), player called "boy" | Character.AI, Tests 5, 6, 7, 10 | Annotate formality in the training data and add a register control (Section 3) |
| Indian festival treated as foreign | Character.AI, Test 9 | Put Maharashtrian lore and culture in the data and in the NPC's persona card |
| Western-fantasy world despite Indian names | Both tools, Tests 1 to 4 | A world card that sets the village, food, festivals and landmarks in Maharashtra |
| Drifted between Hindi, Marathi and English | ChatGPT, Tests 7, 8, 9 | One base language per NPC, with English or Hindi words allowed only as natural insertions |
| Prices and loans changed, story details dropped | ChatGPT, Tests 7, 8, 9 | Keep key facts (name, goal, items, prices) in a small structured memory outside the model |
| Persona dropped after one command | Both tools, Test 11 | Stronger persona instructions and a separate rule for out-of-world questions |

*Table 1: Observed problems and the fix suggested for each*

## 2. (a) Required Linguistic Resources

Table 2 lists what I think is needed. The tool and corpus names are the ones I know of for Marathi and Indian languages. I have not tested them myself, so versions, licences and quality would need to be checked before use.

| Resource | Options I would look at | Why the NPC needs it |
|---|---|---|
| Text corpora | L3Cube-MahaCorpus, IndicCorp (AI4Bharat), Marathi Wikipedia | General Marathi vocabulary, grammar and everyday phrasing for the base model |
| Game dialogue data | A small hand-built set (I know of no large Marathi one) | Teaches the NPC speech style, narration and quest talk, which news and Wikipedia text cannot |

| Resource | Options I would look at | Why the NPC needs it |
|---|---|---|
| Language models | MahaBERT, IndicBERT, MuRIL for understanding. A multilingual LLM such as Llama or Gemma, fine-tuned on Marathi, for generation | Understanding the player's input, and writing fluent in-character replies |
| Tokenizer | One with good Devanagari coverage | If Marathi words are split into many small pieces, replies get slower, costlier and less fluent |
| POS tagger and morphology | Indic NLP Library, IIT Bombay CFILT Marathi tools | Marathi attaches endings to words (see below), so the system needs to split and tag them |
| Marathi WordNet | IIT Bombay CFILT Marathi WordNet | Picking the right sense of a word, which connects to my Assignment 2 |
| Language ID and transliteration | Indic NLP Library, AI4Bharat transliteration tools | Telling Hindi from Marathi, and turning romanized Marathi into Devanagari |
| Translation | IndicTrans2 | A fallback, and a way to bootstrap training data from English RPG dialogue |
| Speech (optional) | Marathi speech recognition and text-to-speech, for example from AI4Bharat | Only if the NPCs should be heard and the player should speak |

*Table 2: Linguistic resources for a Marathi game dialogue system*

The morphology point is visible in my own test input. In Test 6, "gavatil bajaracha rasta" is built from gav (village) plus "atil" (in the), and bajar (market) plus "acha" (of the). In Test 10, "tumchi tabyet" uses "tumchi" because tabyet is feminine, and it would be "tumcha" with a masculine noun. A system that treats each surface form as a separate word will see far more unknown words in Marathi than in English, which is why a morphological analyzer and a good tokenizer matter.

## Concrete steps to adapt the tool

1. **Collect and clean Marathi text.** Gather MahaCorpus, IndicCorp and Wikipedia text. Normalise Unicode so the same Devanagari word is not stored in two forms, and remove duplicates and noisy lines.
2. **Choose the base model.** Use an encoder such as MahaBERT or MuRIL for understanding tasks (language ID, intent), and a multilingual LLM for generating replies. Check how well each handles Marathi before committing.
3. **Build a Marathi game dialogue dataset.** Aim for at least 5,000 NPC conversations in three kinds: pure Marathi, Marathi with English words, and Marathi with Hindi words. Tag each with the speaker's role, social status, formality and scene. Write a core set by hand with native speakers, and bootstrap more by translating English RPG dialogue with IndicTrans2 and then correcting it by hand.
4. **Fine-tune.** Train the base LLM on this data using a light method such as LoRA, so that it keeps its English ability and learns the Marathi NPC style.
5. **Add a language layer.** Detect the language and script of every player message. By default reply in the same language and the same script, so a player who types romanized Marathi gets romanized Marathi back.
6. **Write the persona and world in Marathi.** Give each NPC a card with name, dialect, status, relationship to the player and a Maharashtrian world (village, wada, temple, bazaar, festivals, food). Add a short structured memory for the player's name, goal, items and agreed prices, so these cannot drift.
7. **Control the register.** Choose between "tu", "tumhi" and a more formal form using the NPC's status and the politeness of the player's message (Section 3).
8. **Add safety.** Build a list of Marathi abusive words and slang, including romanized spellings, and filter or handle them. Add a rule that an out-of-world question is answered in character, and that a real question about being an AI gets an honest, short answer.
9. **Evaluate with native speakers.** Rate fluency, grammar, politeness, cultural accuracy and immersion on a 1 to 5 scale. Re-run my 11 tests in Marathi as a regression check.

## 3. (b) Socio-linguistic Challenges

**Code-switching.** Real Marathi speakers mix Marathi with Hindi and English inside one sentence. My own inputs did this: Test 7 was Hinglish, and Test 10 mixed Hindi and Marathi in one sentence. ChatGPT also mixed Hindi and English in Test 9. A Marathi NPC should be able to say "price kiti aahe?" without sounding odd. The challenge is to allow natural mixing without drifting into a different language, which is what ChatGPT did in Tests 7 and 8. I would keep Marathi as the base language and allow English or Hindi only as inserted words, and include mixed sentences in the training and test data.

**Dialects.** Marathi changes from region to region. Examples are standard Pune Marathi, Varhadi in Vidarbha, Ahirani in Khandesh, Malvani on the Konkan coast, Konkani-influenced speech, and the Mumbai mix with Hindi and English. Public text is mostly standard Marathi, so dialect data is scarce. I would start with standard Marathi plus one or two dialects, store the dialect in the NPC card, and have native speakers review it so that dialects do not turn into caricatures.

**Script variation.** Players type Devanagari, but just as often romanized Marathi, and spellings are not fixed: "kay", "kaay" and "kai" can all mean the same word. In my tests the Devanagari inputs (Tests 5 and 6) and the romanized ones (Tests 7 and 10) were handled differently by the tools. ChatGPT also answered my romanized inputs in Devanagari, which a player who types in Roman letters may not want. The fix is to convert romanized input to Devanagari internally, match words loosely, include both scripts in the training data, and reply in the script the player used.

**Honorifics and formality.** Marathi separates "tu" (familiar), "tumhi" (respectful) and more formal forms, and uses kinship words such as Kaka, Dada, Tai and Mama for people who are not relatives. Character.AI ignored all of this, and ChatGPT used "tu" for a player who had used "tumhi". I would give each NPC a status and a default register (for example a village elder, a shopkeeper, a noble), and let the player's politeness adjust it. An elder should be answered with respect, while a gruff blacksmith can still use "tu" with a younger player if the story fits.

**Cultural grounding.** The NPC should know what the player's culture takes for granted. In Test 9, Character.AI did not know Ganesh Chaturthi, aarti or modak, and both tools defaulted to a Western-fantasy world. The persona and data should include Maharashtrian festivals (Ganesh Chaturthi, Gudi Padwa, Diwali), food (modak, puran poli, vada pav), proverbs, forts and village life, in place of taverns, wolves and bandits. Cultural errors are easy to make, so a native speaker should review this part.

## 4. A Quick Prompt-Level Fix

Fine-tuning takes time, but part of the problem can be reduced with a better setup prompt. In Part (ii) my ChatGPT prompt said nothing about language or culture. A prompt like the one below would address the main failures directly. I have not tested it, and I suggest re-running Tests 5 to 10 with it to see what changes.

> *"You are Bhairav, a blacksmith in a village in Maharashtra. Reply in the language and script the player uses. If the player mixes languages, keep Marathi as your base and use English or Hindi words only when a real Marathi speaker would. Match the player's politeness: use tumhi with respectful speakers and elders. Know local festivals, food and places. Keep a note of the player's name, goal, items and agreed prices, and never contradict them. Stay in character unless the player sincerely asks whether you are an AI."*

## 5. How I Would Check It Worked

| What to check | How | Success looks like |
|---|---|---|
| Reply language and script | Re-run Tests 5, 6, 7 and 10 in Marathi, Devanagari and romanized | Reply in the player's language and script every time |
| Politeness | Inputs with tu, tumhi, Kaka and Bhai, judged by native speakers | Register matches the NPC's status and the player's tone |

| What to check | How | Success looks like |
|---|---|---|
| Cultural accuracy | Test 9 style inputs on festivals, food and places | Recognised, with no wrong facts |
| Memory and consistency | Repeat the price and loan sequence from Tests 3 to 9 | No price changes, no forgotten story details |
| Persona strength | Test 11 style commands and off-topic questions | Stays in character, honest when sincerely asked |

*Table 3: Evaluation plan*

## 6. Conclusion

My tests showed that the main gap is not understanding. Both tools understood Hindi and Marathi. The gap is in what the NPC says back: the language, the politeness, the culture and the consistency. Adapting a game dialogue tool for Marathi therefore needs more than a translation layer. It needs Marathi dialogue data, tools that handle Marathi word forms, control over register and script, a Maharashtrian world to speak about, and native speakers to check the results.